Exam-style induction problems

Answers and worked solutions · Day 12 · Wed 4 Nov · pack M9

5. Practice

From your course files

PQ 7.11 (practice questions).

Ratings: 2.2 kW, 50 Hz, 1420 rpm, 415 V, 4.9 A, $\eta = 0.80$, pf 0.78.

a. $p = 3000/1420 = 2.11$, so $p = 2$. $n_S = 1500$ rpm.

b. $\omega = 2\pi \times 1420/60 = 148.70$ rad/s. $\tau = 2200/148.70 =$ **14.80 Nm**.

(c)(i) $P_{IN} = \sqrt{3} V_L I_L \cos\phi = \sqrt{3} \times 415 \times 4.9 \times 0.78 =$ **2.747 kW**. (ii) $P_{IN} = P_{OUT}/\eta = 2200/0.80 =$ **2.750 kW**. They agree, which shows the rating plate is self-consistent.

d. $s = 80/1500 = 0.05333$. Ideal efficiency $= 1 - s =$ **94.67%**. The actual 80% is lower because the ideal value only counts rotor copper loss. Stator copper loss, iron loss and friction and windage are extra.

Course answers: 1500 rpm, 14.80 Nm, 2.747 kW, 2.750 kW, 94.67%. All match.

PQ 7.16 (practice questions).

Constant V/f keeps the flux constant: $V = 460 \times f/60$.

- 30 Hz: $V = 460 \times 30/60 =$ **230 V**
- 50 Hz: $V = 460 \times 50/60 =$ **383.3 V**

Course answers: 230 V, 383.3 V.

PQ 7.12 (practice questions; do (c)–(d) here).

415 V, 1455 rpm, 50 Hz, star. From part (a) (method in M8): $R_1 = 3\ \Omega$. Blocked rotor per phase: 37.53 V, 4.8 A, 116.7 W, so $R = 5.064\ \Omega$, $R_2' = 2.064\ \Omega$, $X = 5.957\ \Omega$. No-load per phase: 239.6 V, 3.0 A, 68.33 W, so $R_C = 840.1\ \Omega$, $X_M = 80.23\ \Omega$.

(c) At rated speed. $p = 2$, $n_S = 1500$ rpm, $s = 45/1500 = 0.03$, $R_2'/s = 68.79\ \Omega$.

$$Z_1 = (3 + 68.79) + j5.957 = 71.79 + j5.957 = 72.04\angle 4.743^\circ\ \Omega$$
$$I_2' = 239.6/72.04\angle 4.743^\circ = 3.326\angle -4.743^\circ = 3.315 - j0.2751\ \text{A}$$
$$I_M = 239.6/840.1 - j\,239.6/80.23 = 0.2852 - j2.986\ \text{A}$$
$$I_1 = 3.600 - j3.261 = \mathbf{4.858\angle -42.18^\circ\ A}$$

- pf $= \cos 42.18^\circ =$ **0.7411 lag**
- $P_{IN} = 3 \times 239.6 \times 4.858 \times 0.7411 =$ **2.588 kW**
- $P_{AG} = 3 \times 3.326^2 \times 68.79 = 2283$ W, $\tau = 2283/157.08 =$ **14.53 Nm**
- $\omega = 2\pi \times 1455/60 = 152.37$ rad/s, $P_{OUT} = 14.53 \times 152.37 =$ **2.215 kW**
- $\eta = 2.215/2.588 =$ **85.58%**

(d) Losses. Copper $= 3 \times 3.326^2 \times (3 + 2.064) =$ **168.1 W**. Iron $= 3 \times 239.6^2/840.1 = 205.0$ W (equals the no-load input power, as expected). Total **373.1 W**. $\eta = 2215/(2215 + 373.1) =$ **85.58%**, the same as (c).

Course answers: $I_2' = 3.323\angle -4.738^\circ$ A, 4.855 A, 0.7408 lag, 2.585 kW, 14.52 Nm, 2.212 kW, 85.58%. Copper 167.8 W, iron 205.0 W, total 372.8 W. Mine agree to within 0.1%. The small difference comes from rounding of the part (a) parameters. The efficiency matches exactly.

PQ 7.14 (practice questions; do (a)–(d) and (f) here).

415 V star, 50 Hz, 2790 rpm. $R_1 = 0.2$, $R_2' = 0.16$, $X = 0.3 + 0.3 = 0.6\ \Omega$, $X_M = 12\ \Omega$, no R_C. $p = 1$, $n_S = 3000$ rpm, $\omega_S = 314.16$ rad/s, $V_1 = 239.6$ V.

(a) Locked rotor, $s = 1$, ignore X_M. $Z = 0.36 + j0.6 = 0.6997\angle 59.04^\circ\ \Omega$.

- $I = 239.6/0.6997 =$ **342.4 A**
- $P_{IN} = 3 \times 342.4^2 \times 0.36 =$ **126.6 kW** (all of it is copper loss)
- $\tau = 3 \times 342.4^2 \times 0.16/314.16 =$ **179.2 Nm** (P_{AG} uses only R_2', since $s = 1$)

(b) Rated speed. $s = 210/3000 = 0.07$, $R_2'/s = 2.286\ \Omega$.

$$Z_1 = 2.486 + j0.6 = 2.557\angle 13.57^\circ\ \Omega, \quad I_2' = 93.70\angle -13.57^\circ = 91.08 - j21.99\ \text{A}$$
$$I_M = 239.6/j12 = -j19.97\ \text{A}$$
$$I_1 = 91.08 - j41.95 = \mathbf{100.3\angle -24.73^\circ\ A}$$

pf $= \cos 24.73^\circ =$ **0.9083 lag**. $P_{IN} = 3 \times 239.6 \times 100.3 \times 0.9083 =$ **65.47 kW**.

(c) $P_{AG} = 3 \times 93.70^2 \times 2.286 = 60.20$ kW. $\tau = 60.20\text{k}/314.16 =$ **191.6 Nm**. $\omega = 292.17$ rad/s, $P_{OUT} = 191.6 \times 292.17 =$ **55.99 kW**. $\eta = 55.99/65.47 =$ **85.52%**.

(d) $\tau_{start}/\tau_R = 179.2/191.6 = 0.935$. $I_{start}/I_R = 342.4/100.3 = 3.414$.

(f) No-load ($s \to 0$, $R_2'/s \to \infty$, $I_2' = 0$). $I_1 = I_M = 239.6/12 =$ **19.97 A**, which is $19.97/100.3 = 0.1991$ of rated current.

(e) is the maximum-torque method (M8). Check answers: 2241 rpm, 329.3 Nm, 1.719 × rated. **(g)** is the sketch method in M8, using these three torques.

Course answers: 342.4 A, 126.6 kW, 179.2 Nm; 100.3 A, 0.9083 lag, 65.47 kW; 191.6 Nm, 55.99 kW, 85.52%; 0.9353, 3.414; 19.97 A, 0.1991. All match, except τ_{start}/τ_R: unrounded values give 0.9349. The course's 0.9353 comes from dividing the rounded torques 179.2/191.6, so either is fine.

PQ 7.15 (practice questions; do (b)(i), (ii), (iv) here).

460 V, 60 Hz, 1760 rpm, star. $R_1 = 0.25$, $R_2' = 0.2$, $X = 1.2 + 1.1 = 2.3\ \Omega$. $V_1 = 460/\sqrt{3} = 265.6$ V. $p = 3600/1760 = 2.05$, so $p = 2$, $n_S = 1800$ rpm, $\omega_S = 188.50$ rad/s.

- i. Synchronous: $s = 0$, $R_2'/s \to \infty$, so **τ = 0 Nm**.
- ii. Rated: $s = 40/1800 = 0.02222$, $R_2'/s = 9.0\ \Omega$.
$$\tau = \frac{3 \times 265.6^2 \times 9}{188.50\left[9.25^2 + 2.3^2\right]} = \textbf{111.2 Nm}.$$
- iv. Locked: $s = 1$. $\tau = \dfrac{3 \times 265.6^2 \times 0.2}{188.50\left[0.45^2 + 2.3^2\right]} = \textbf{40.88 Nm}$.

iii. and (d) use $s_{max} = R_2'/|R_1 + jX| = 0.08645$ (M8): 1644 rpm, 218.9 Nm, and $R_2' = 2.314\ \Omega$ for maximum torque at standstill. All match the course answers.

New questions

New Q1 (15 marks, easy–medium).

(a) 6 poles, $p = 3$: $n_S = 3000/3 = \textbf{1000 rpm}$, $\omega_S = 104.72$ rad/s. $s = 30/1000 = \textbf{0.03}$. $f_2 = sf = \textbf{1.5 Hz}$. $R_2'/s = 0.6/0.03 = \textbf{20 Ω}$. [3]

(b) Delta: $V_1 = V_L = 400$ V. [1]

$$Z_1 = (0.8 + 20) + j3 = 20.8 + j3 = 21.02\angle 8.207^\circ\ \Omega$$
$$I_2' = 400/21.02\angle 8.207^\circ = 19.03\angle -8.207^\circ = 18.84 - j2.717\ \text{A}$$
$$I_M = 400/300 - j\,400/60 = 1.333 - j6.667\ \text{A}$$
$$I_1 = 20.17 - j9.384 = \mathbf{22.25\angle -24.95^\circ\ A\ (phase)}$$

Line current $= \sqrt{3} \times 22.25 = \textbf{38.53 A}$. pf $= \cos 24.95^\circ = \textbf{0.9067 lag}$. [4]

(c)

- $P_{IN} = 3 \times 400 \times 22.25 \times 0.9067 = \textbf{24.21 kW}$ [1]
- $P_{AG} = 3 \times 19.03^2 \times 20 = \textbf{21.74 kW}$ [1]
- $\tau = 21.74\text{k}/104.72 = \textbf{207.6 Nm}$ [1]
- $\omega = 2\pi \times 970/60 = 101.58$ rad/s, $P_{OUT} = 207.6 \times 101.58 = \textbf{21.09 kW}$ [1]

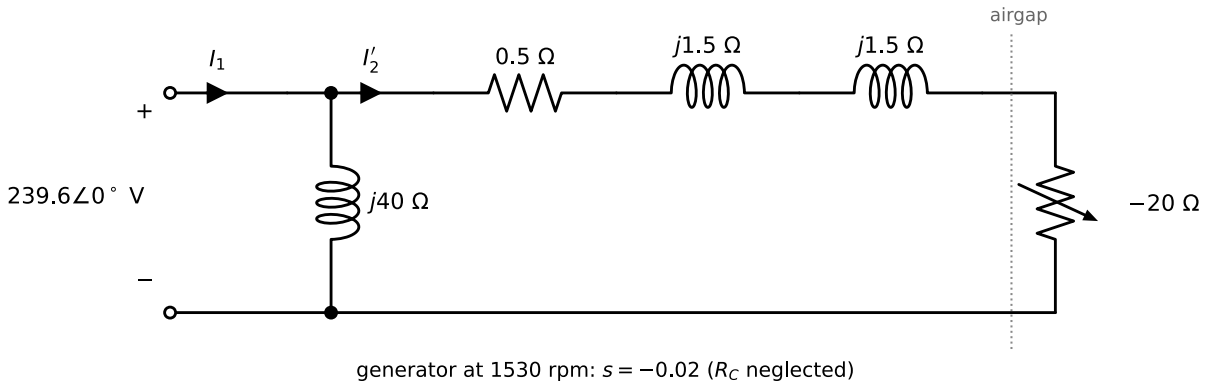
- $\eta = 21.09/24.21 = \mathbf{87.10\%}$ [1]

(d) Stator copper $3 \times 19.03^2 \times 0.8 = 869.5$ W. Rotor copper $3 \times 19.03^2 \times 0.6 = 652.1$ W ($= sP_{AG} = 0.03 \times 21737$ (checks)). Iron $3 \times 400^2/300 = 1600$ W. Total 3122 W. $24.21 - 3.12 = 21.09$ kW $= P_{OUT}$. [2]

New Q2 (17 marks, medium).

(a) $n_S = 1500$ rpm. $n > n_S$, so it is **generating**. $s = (1500 - 1530)/1500 = \mathbf{-0.02}$. $R_2'/s = 0.4/(-0.02) = \mathbf{-20\ \Omega}$. [3]

(b) $V_1 = 239.6$ V.



generator at 1530 rpm: $s = -0.02$ (R_C neglected)

New Q2 generator circuit: 239.6 V across j40 ohm, then 0.5 ohm, j1.5 ohm, j1.5 ohm and R2′/s = -20 ohm

$$Z_1 = (0.5 - 20) + j3 = -19.5 + j3 = 19.73\angle 171.3^\circ\ \Omega$$
$$I_2' = 239.6/19.73\angle 171.3^\circ = \mathbf{12.14\angle -171.3^\circ\ A} = -12.00 - j1.847\ \text{A}$$
$$I_M = 239.6/j40 = -j5.990\ \text{A}$$
$$I_1 = -12.00 - j7.837 = \mathbf{14.33\angle -146.9^\circ\ A}$$

[4]

Using $S = 3V_1 I_1^*$: $P = 3 \times 239.6 \times 14.33 \cos(-146.9^\circ) = \mathbf{-8.628\ kW}$, $Q = 3 \times 239.6 \times 14.33 \sin(146.9^\circ) = \mathbf{+5.633\ kVAr}$. $P < 0$: 8.628 kW is delivered **to** the grid. $Q > 0$: it still **absorbs** 5.633 kVAr from the grid. [2]

(c)

- $P_{AG} = 3 \times 12.14^2 \times (-20) = \mathbf{-8.849\ kW}$ [1]
- $\tau = -8849/157.08 = \mathbf{-56.33\ Nm}$ (the turbine must apply 56.33 Nm) [1]
- $\omega = 2\pi \times 1530/60 = 160.22$ rad/s, $P_{OUT} = \tau\omega = -9.026$ kW, so the mechanical input is **9.026 kW** [1]
- $\eta_{gen} = 8.628/9.026 = \mathbf{95.59\%}$ [1]
- Check: copper loss $3 \times 12.14^2 \times 0.9 = 398.2$ W $= 9026 - 8628$ (checks) [1]

(d) The rotor has no separate excitation. The rotating flux is set up only by the stator magnetising current through X_M, which comes from the supply. So the machine always draws lagging reactive power (V_1^2/X_M per phase, plus the leakage VArs), whether motoring or generating. The grid (or local capacitors) must provide these VArs. A stand-alone induction generator needs a capacitor bank. [3]

New Q3 (18 marks, medium–hard).

$V_1 = 239.6$ V, $n_S = 1500$ rpm, $\omega_S = 157.08$ rad/s, $X = 2.4\ \Omega$, $3V_1^2 = 172{,}225\ \text{V}^2$.

(a) $s = 30/1500 = 0.02$, $R_2'/s = 20\ \Omega$. $\tau = \dfrac{172225 \times 20}{157.08\left[20.5^2 + 2.4^2\right]} =$ **51.47 Nm**.
$|I_1| = 239.6/|20.5 + j2.4| =$ **11.61 A**. [3]

(b) Let $x = R_2'/s$. Rearranging the torque formula with $\tau_L = 40$ Nm:

$$\tau_L\omega_S x^2 + (2R_1\tau_L\omega_S - 3V_1^2)x + \tau_L\omega_S(R_1^2 + X^2) = 0$$
$$6283.2\,x^2 - 165942\,x + 37762 = 0$$

[2]

Roots: $x = 26.18\ \Omega$ or $x = 0.2296\ \Omega$. [1]

These give $s = 0.4/26.18 = 0.01528$ or $s = 0.4/0.2296 = 1.742$. The second is outside $0 < s \leq 1$ (it is braking, with the rotor turning backwards). Choose the larger x, which is the small slip on the stable, steep part of the curve. [1]

$s =$ **0.01528**, $n = 1500(1 - 0.01528) =$ **1477.1 rpm**. [2]

(c) Linear: $s \approx 0.02 \times 40/51.47 = 0.01554$, $n \approx$ **1476.7 rpm**. [2] The error is only 0.4 rpm, because both points are at low slip where $\tau \propto s$ (Lecture 7.8.1). This approximation gets worse near maximum torque. [1]

(d) With $V_1 = 0.9 \times 239.6$, $3V_1^2$ falls by 0.81 to 139,502. The new quadratic is $6283.2x^2 - 133219x + 37762 = 0$, giving $x = 20.92\ \Omega$, $s = 0.01912$, $n =$ **1471.3 rpm**. [3] Estimate: at low slip $\tau \propto V^2 s$, so $s \approx 0.01528/0.81 = 0.01886$, $n \approx 1471.7$ rpm. Close. [1]

(e) $s = 1$: $\tau_{st} = \dfrac{172225 \times 0.4}{157.08\left[0.9^2 + 2.4^2\right]} =$ **66.75 Nm** > 40 Nm, so it **can** start the load.
$I_{st} = 239.6/|0.9 + j2.4| = 93.48$ A. Running current at 40 Nm:
$239.6/|0.5 + 26.18 + j2.4| = 8.944$ A. Ratio $= 10.45$. [2]

New Q4 (20 marks, hard: a full Q3-style chain).

(a) Blocked rotor, per phase: $V = 75/\sqrt{3} = 43.30$ V, $I = 8$ A, $P = 173.3$ W.
$R = 173.3/8^2 = 2.708\ \Omega$, so $R_2' = 2.708 - 1.2 =$ **1.508 Ω**. $S = 346.4$ VA,

$Q = \sqrt{346.4^2 - 173.3^2} = 299.9$ VAr, $X = 299.9/64 =$ **4.686 Ω**. [2]

No load, per phase: $V = 239.6$ V, $I = 4.0$ A, $P = 86.67$ W. $R_C = 239.6^2/86.67 =$ **662.4 Ω**. $S = 958.4$ VA, $Q = 954.5$ VAr, $X_M = 239.6^2/954.5 =$ **60.15 Ω**. [2]

(b) $p = 3000/1440 = 2.08$, so $p = 2$: **4 poles**, $n_S =$ **1500 rpm**, $s = 60/1500 =$ **0.04**. Ideal efficiency $1 - s =$ **96%**. [3]

(c) $R_2'/s = 1.508/0.04 = 37.71\ \Omega$.

> New Q4 circuit at rated speed: 239.6 V, 662.4 ohm parallel j60.15 ohm, then 1.2 ohm, j4.686 ohm and 37.71 ohm

$$Z_1 = (1.2 + 37.71) + j4.686 = 38.91 + j4.686 = 39.19\angle 6.868^\circ\ \Omega$$
$$I_2' = 239.6/39.19\angle 6.868^\circ = 6.114\angle -6.868^\circ = 6.070 - j0.7311\ \text{A}$$
$$I_M = 239.6/662.4 - j\,239.6/60.15 = 0.3617 - j3.984\ \text{A}$$
$$I_1 = 6.432 - j4.715 = \mathbf{7.975\angle -36.24^\circ\ A}$$

[3]

- pf $= \cos 36.24^\circ =$ **0.8065 lag** [1]
- $P_{IN} = 3 \times 239.6 \times 7.975 \times 0.8065 =$ **4.623 kW** [1]
- $P_{AG} = 3 \times 6.114^2 \times 37.71 = 4229$ W, $\tau = 4229/157.08 =$ **26.92 Nm** [1]
- $P_{OUT} = 26.92 \times 150.80 =$ **4.059 kW**, $\eta = 4.059/4.623 =$ **87.81%** [1]

(Check: $P_{CU} = 3 \times 6.114^2 \times 2.708 = 303.7$ W, $P_{FE} = 260$ W,
$4623 - 304 - 260 = 4059$ W (checks).)

(d) $s = (1500 - 1560)/1500 = -0.04$, $R_2'/s = -37.71\ \Omega$. $Z_1 = -36.51 + j4.686$,
$I_2' = 6.509\angle -172.7^\circ$ A $= -6.457 - j0.8288$ A.
$I_1 = I_2' + I_M = -6.095 - j4.812 = 7.766\angle -141.7^\circ$ A. [1]

- $P_{AG} = 3 \times 6.509^2 \times (-37.71) = -4793$ W, $\tau = -4793/157.08 =$ **−30.52 Nm**. So **30.52 Nm** must be applied. [1]
- $P_{IN} = 3 \times 239.6 \times 7.766 \cos(-141.7^\circ) = -4.381$ kW. So **4.381 kW** is delivered to the supply. [1]
- Mechanical input $= 30.52 \times (2\pi \times 1560/60) = 30.52 \times 163.36 = 4.985$ kW. $\eta_{gen} = 4.381/4.985 =$ **87.88%**. [1]

(e) The ideal efficiency $1 - s$ only accounts for rotor copper loss (sP_{AG}). The real machine also has stator copper loss (here $3I_2'^2R_1 = 134.6$ W) and iron loss (260 W), plus friction and windage, which the model does not include. [2]

6. Quick recall check

Question

1. Star: $415/\sqrt{3} = 239.6$ V. Delta: 415 V.
2. Divide by ω_S (synchronous). Multiply by ω (actual rotor speed).
3. s, R_2'/s, P_{IN} and τ are negative. Q_{IN} stays positive, because the machine still needs magnetising VArs.
4. Rotor copper loss $= sP_{AG}$. Mechanical power $= (1-s)P_{AG}$.
5. The larger R_2'/s (smallest slip). It is on the stable, steep part of the curve near n_S.